

# Chapter 17: Differentiability of Convex Functions

## Subdifferential Examples & Properties

Convex Analysis and Monotone Operator Theory

Pages 533–565

# Chapter Overview

- **Main Topic:** Differentiability and subdifferentiability of convex functions
- **Key Concepts:**
  - Directional derivatives
  - Gâteaux and Fréchet differentiability
  - Subdifferentials and subgradients
  - Characterizations of convexity
- **Applications:**
  - Optimization theory
  - Monotone operator theory
  - Numerical algorithms

# Directional Derivatives (Section 17.1)

## Definition (Directional Derivative)

For  $f : \mathcal{H} \rightarrow \mathbb{R} \cup \{+\infty\}$  proper and convex, with  $x \in \text{dom} f$  and  $y \in \mathcal{H}$ :

$$f'(x; y) = \inf_{\alpha \in \mathbb{R}_{++}} \frac{f(x + \alpha y) - f(x)}{\alpha}$$

## Proposition (Proposition 17.2)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper and convex. Then:

- (i)  $f'(x; y)$  exists in  $[-\infty, +\infty]$
- (ii)  $f'(x; y - x) + f(x) \leq f(y)$
- (iii)  $\nabla f$  is sublinear if  $f$  is differentiable
- (iv)  $f'(x; 0) = 0$  and  $f'(x; \cdot)$  is sublinear

# Key Properties of Directional Derivatives

## Proposition (Proposition 17.3)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper and convex, and  $x \in \text{dom} f$ . Then:

$$x \in \text{Argmin} f \iff f'(x; \cdot) \geq 0$$

## Proposition (Proposition 17.4)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be convex and  $x \in \mathcal{H}$ . If  $f$  is Gâteaux differentiable at  $x$ , then:

$$x \in \text{Argmin} f \iff \nabla f(x) = 0$$

- The directional derivative characterizes whether a point is a minimum
- Gâteaux differentiability provides a linear characterization

# Convexity Characterizations (Section 17.2)

## Proposition (Proposition 17.7)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper with open convex domain. If  $f$  is Gâteaux differentiable on  $\text{dom} f$ , then the following are equivalent:

- (i)  $f$  is convex
- (ii)  $\langle y - x \mid \nabla f(x) \rangle + f(x) \leq f(y)$  for all  $x, y$
- (iii)  $\nabla f$  is monotone

If  $f$  is twice Gâteaux differentiable, then equivalently:

- (iv)  $\langle z \mid \nabla^2 f(x) z \rangle \geq 0$  for all  $z$

# First-Order Characterization Example

## Example (Example 17.8)

Let  $A \in \mathcal{B}(\mathcal{H})$  and  $f : \mathcal{H} \rightarrow \mathbb{R} : x \mapsto \langle x \mid Ax \rangle$ . Then  $f$  is convex if and only if:

$$\langle x \mid (A + A^*)x \rangle \geq 0 \quad \forall x \in \mathcal{H}$$

## Example (Example 17.9)

Given  $h : \mathcal{H} \rightarrow [-\infty, +\infty]$  Fréchet differentiable on open convex domain  $D$ , define:

$$f(x) = \begin{cases} h(x), & x \in D \\ \lim_{\alpha \downarrow 0} h((1 - \alpha)y + \alpha x), & x \in \text{bdry } D \\ +\infty, & x \notin \overline{D} \end{cases}$$

Then  $f \in \Gamma_0(\mathcal{H})$  and  $f|_D = h|_D$  with lower semicontinuity.

# Strict Convexity Characterizations (Section 17.3)

## Proposition (Proposition 17.10)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper with open convex domain. If  $f$  is Gâteaux differentiable on  $\text{dom} f$ , then these are equivalent:

- (i)  $f$  is strictly convex
- (ii)  $x \neq y \Rightarrow \langle y - x \mid \nabla f(x) \rangle + f(x) < f(y)$
- (iii)  $x \neq y \Rightarrow \langle x - y \mid \nabla f(x) - \nabla f(y) \rangle > 0$

If  $f$  is twice Gâteaux differentiable:

- (iv)  $z \neq 0 \Rightarrow \langle z \mid \nabla^2 f(x)z \rangle > 0$  for all  $z \in \mathcal{H}$

# Strict Convexity Examples

## Example (Example 17.11)

Let  $A \in \mathcal{B}(\mathcal{H})$  and  $f : \mathcal{H} \rightarrow \mathbb{R} : x \mapsto \langle x \mid Ax \rangle$ . Then  $f$  is strictly convex if and only if  $A + A^*$  is strictly monotone.

## Example (Example 17.13)

The function

$$h : \mathbb{R}^2 \rightarrow [-\infty, +\infty] : (\xi, \eta) \mapsto \begin{cases} \eta^2 + \eta^2/\xi, & \xi > 0, \eta > 0 \\ +\infty, & \text{otherwise} \end{cases}$$

is strictly convex in the interior of its domain.

## Definition and Basic Properties (Section 17.4)

### Definition (Descent Direction)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper and convex, with  $x \in \text{dom} f$ . Then  $y \in \mathcal{H}$  is a descent direction of  $f$  at  $x$  if there exists  $\varepsilon \in \mathbb{R}_{++}$  such that:

$$\forall \alpha \in [0, \varepsilon] \quad f(x + \alpha y) < f(x)$$

### Proposition (Proposition 17.21)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper and convex. Then:

- (i)  $y$  is a descent direction at  $x$  iff  $f'(x; y) < 0$
- (ii) If  $f$  is differentiable at  $x \notin \text{Argmin} f$ , then  $-\nabla f(x)$  is a descent direction

# Steepest Descent Direction

## Proposition (Proposition 17.22 – Steepest Descent)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper and convex with  $x \in (\text{cont } f) \setminus (\text{Argmin } f)$ . Set  $u = P_{\partial f(x)}(0)$  and  $z = -u/\|u\|$ . Then  $z$  is the unique minimizer of  $f'(x; \cdot)$  over  $B(0; 1)$ , satisfying:

$$f'(x; z) = \max_{z \in \partial f(x)} \langle z \mid \partial f(x) \rangle = -\|u\|$$

- The steepest descent direction is the negative of the projection onto the subdifferential
- This characterizes the direction of maximum decrease

# Chebyshev Center Example

## Proposition (Proposition 17.25 – Chebyshev Center)

Let  $C$  be a nonempty compact convex subset of  $\mathcal{H}$ , and define:

$$f : \mathcal{H} \rightarrow \mathbb{R} : x \mapsto \max_{y \in C} \|x - y\|^2$$

Then:

- (i)  $f$  is continuous, strongly convex, and supercoercive
- (ii)  $\text{dom } \partial f = \mathcal{H}$ ,  $\text{gra } \partial f$  is closed, and the sets  $(\partial f(x))_{x \in \mathcal{H}}$  are compact
- (iv)  $\partial f(x) = 2(x - \text{conv } \partial_C(x))$
- (v)  $f$  has a unique minimizer  $r$ , the Chebyshev center, characterized by  $r \in \text{conv } \partial_C(r)$

# Convexity from Directional Derivatives (Section 17.5)

## Proposition (Proposition 17.27)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper with  $x_0 \in \text{dom} f$  and  $x_1 \in \mathcal{H}$  such that  $[x_0, x_1] \subset \text{dom} f$ . Suppose that  $f|_{[x_0, x_1]}$  is lower semicontinuous and that for every  $x \in [x_0, x_1]$ ,  $f'(x; x_1 - x_0) \leq 0$ . Then  $f(x_1) \leq f(x_0)$ .

## Proposition (Proposition 17.29)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper with  $\text{dom} f$  convex and  $f|_{\text{dom} f}$  lower semicontinuous. Suppose that for every  $x, y \in \text{dom} f$ :

$$f'(x; y - x) \leq -f'(y; x - y)$$

Then  $f$  is convex.

# Directional Derivatives and Convexity (continued)

## Corollary (Corollary 17.30)

Let  $f : \mathbb{R} \rightarrow [-\infty, +\infty]$  be proper with  $I = \text{dom} f$  an interval. Suppose that  $f|_I$  is lower semicontinuous and that for every  $x, y \in I$  with  $x < y$ :

- (i)  $f'_-(x)$  and  $f'_-(y)$  exist
- (ii)  $f'_-(x) \leq f'_-(y)$  (monotonicity)

Then  $f$  is convex.

This is the univariate analogue of Proposition 17.29.

# Differentiability Concepts (Section 17.6)

## Definition (Gâteaux and Fréchet Differentiability)

Let  $f : \mathcal{H} \rightarrow \mathbb{R} \cup \{+\infty\}$  be proper and convex at  $x$ :

- **Gâteaux differentiable:**  $f'(x; y) = \langle y \mid \nabla f(x) \rangle$  for some  $\nabla f(x) \in \mathcal{H}$
- **Fréchet differentiable:**  $\lim_{y \rightarrow 0} \frac{f(x+y) - f(x) - \langle y \mid \nabla f(x) \rangle}{\|y\|} = 0$

## Proposition (Proposition 17.31)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be proper and convex, and  $x \in \text{dom} f$ . Then:

- (i) If  $f$  is Gâteaux differentiable at  $x$ , then  $\partial f(x) = \{\nabla f(x)\}$
- (ii) If  $x \in \text{cont} f$  and  $\partial f(x)$  is a singleton  $u$ , then  $f$  is Gâteaux differentiable and  $\nabla f(x) = u$

# Chain Rule for Differentiable Functions

## Proposition (Proposition 17.32)

Let  $f : \mathcal{H} \rightarrow \mathbb{R}$  be continuous and convex, and  $\phi : \mathbb{R} \rightarrow \mathbb{R}$  be convex and differentiable with  $\phi' \geq 0$ . If  $\phi \circ f$  is Gâteaux differentiable at  $x$ , then:

$$\partial(\phi \circ f)(x) = \phi'(f(x))\partial f(x)$$

## Example (Example 17.33)

Let  $C$  be a closed convex set,  $\phi : \mathbb{R} \rightarrow \mathbb{R}$  convex and differentiable on  $\mathbb{R}_+$  with  $\phi' > 0$  there. For  $x \in \mathcal{H}$ :

$$\partial(\phi \circ d_C)(x) = \begin{cases} \{\phi'(d_C(x))(x - P_C x)\}, & x \notin C \\ N_C \cap B(0; \phi'(0)), & x \in \text{bdry } C \\ \{0\}, & x \in \text{int } C \end{cases}$$

# Gâteaux vs Fréchet Differentiability

## Proposition (Proposition 17.39)

Let  $f \in \Gamma_0(\mathcal{H})$  with  $x \in \text{int dom } f$ . Then the following are equivalent:

- (i)  $f$  is Gâteaux differentiable at  $x$
- (ii) Every selection of  $\partial f$  is strong-to-weak continuous at  $x$
- (iii) There exists a selection of  $\partial f$  that is strong-to-weak continuous at  $x$

## Proposition (Proposition 17.41)

Let  $f \in \Gamma_0(\mathcal{H})$  with  $x \in \text{int dom } f$ . Then the following are equivalent:

- (i)  $f$  is Fréchet differentiable at  $x$
- (ii) Every selection of  $\partial f$  is continuous at  $x$
- (iii) There exists a selection of  $\partial f$  that is continuous at  $x$

# Finite-Dimensional Results

## Corollary (Corollary 17.44)

*If  $\mathcal{H}$  is finite-dimensional, then Gâteaux and Fréchet differentiability are the same for functions in  $\Gamma_0(\mathcal{H})$ .*

## Proposition (Proposition 17.45)

*Assume  $\mathcal{H}$  is finite-dimensional. Let  $f \in \Gamma_0(\mathcal{H})$ ,  $x \in \mathcal{H}$ , and  $u \in \mathcal{H}$ . Then:*

$$\partial f(x) = \{u\} \iff f \text{ is Fréchet differentiable at } x \text{ and } u = \nabla f(x)$$

- In finite dimensions, the notions of differentiability coincide
- A singleton subdifferential characterizes Fréchet differentiability

# Monotone Operators Example

## Proposition (Proposition 17.36)

Let  $A \in \mathcal{B}(\mathcal{H})$  be monotone and self-adjoint. Suppose that  $\text{ran } A$  is closed. Define:

$$q_A : \mathcal{H} \rightarrow \mathbb{R} : x \mapsto \frac{1}{2} \langle x \mid Ax \rangle, \quad q_{A^\dagger} : \mathcal{H} \rightarrow \mathbb{R} : x \mapsto \frac{1}{2} \langle x \mid A^\dagger x \rangle$$

Then:

- (i)  $q_A$  is convex, continuous, Fréchet differentiable, with  $\nabla q_A = A$
- (ii)  $q_{A^\dagger} = q_A \circ A^\dagger = \iota_{\ker A} \square q_{A^\dagger}$
- (iii)  $q_{A^\dagger}^* = \iota_{\text{ran } A} + q_A^\dagger$
- (iv)  $q_{A^\dagger}^* \circ A = q_A$

# Continuity from Differentiability (Section 17.7)

## Example (Example 17.47)

Let  $C$  be the closed cone of Example 8.42 and  $f = \iota_C$ . Then:

- $0 \in (\text{core } C) \setminus (\text{int } C)$
- $\partial f(0) = \{0\}$  is a singleton
- $f$  is Gâteaux differentiable at  $0$  with  $\nabla f(0) = 0$
- $f$  is lower semicontinuous at  $0$
- But  $0 \notin \text{int dom } f$ , so  $f$  is not continuous at  $0$

This shows that Gâteaux differentiability does not imply continuity.

# Main Continuity Results

## Proposition (Proposition 17.48)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be convex and Gâteaux differentiable at  $x \in \text{dom} f$ . Then:

- (i)  $f$  is lower semicontinuous at  $x$
- (ii) If  $\mathcal{H}$  is finite-dimensional, then  $x \in \text{int dom } f$  and  $f$  is continuous on  $\text{int dom } f$

## Proposition (Proposition 17.50)

Let  $f \in \Gamma_0(\mathcal{H})$  be Gâteaux differentiable at  $x \in \text{dom} f$ . Then  $x \in \text{int dom } f$  and  $f$  is continuous on  $\text{int dom } f$ .

## Proposition (Proposition 17.51)

Let  $f : \mathcal{H} \rightarrow [-\infty, +\infty]$  be convex and suppose it is Gâteaux differentiable on some open subset  $U$  of  $\text{dom} f$ . Then  $f$  is continuous on  $\text{int dom } f$ .

# Counterexample: Non-Continuous Gâteaux Differentiable Function

## Example (Example 17.49)

Let  $f$  be the discontinuous linear functional of Example 2.27, and set  $g = |f|$ . Then:

- $g$  is convex
- $g : \mathcal{H} \rightarrow \mathbb{R}$  is continuous at  $0 = \text{dom} g$
- $g$  is Gâteaux differentiable at  $0$
- By Proposition 17.48,  $g$  is lower semicontinuous at  $0$
- However,  $g$  is neither continuous nor Fréchet differentiable at  $0$

This shows that Gâteaux differentiability alone does not guarantee continuity or Fréchet differentiability.

# Python: Computing Subdifferentials

```
# Compute directional derivative numerically
def directional_derivative(f, x, y, eps=1e-8):
    alphas = np.logspace(-8, 0, 100)
    diffs = [(f(x + a*y) - f(x))/a
              for a in alphas]
    return np.min(diffs)

# Example: f(x) = x^2
f = lambda x: x**2
x0, y = 1.0, 2.0
f_prime = directional_derivative(f, x0, y)
print(f"f'({x0}; {y}) = {f_prime:.6f}")
```

## Example: Convex Function Analysis

```
class ConvexFunction:
    def __init__(self, f, grad_f):
        self.f = f
        self.grad_f = grad_f

    def is_monotone_gradient(self, x_vals):
        grads = [self.grad_f(x)
                  for x in x_vals]
        return all(grads[i] <= grads[i+1]
                    for i in range(len(grads)-1))

# Example:  $f(x) = x^2$  is convex
f1 = lambda x: x**2
grad_f1 = lambda x: 2*x
func1 = ConvexFunction(f1, grad_f1)
print(func1.is_monotone_gradient(
    np.linspace(-2, 2, 100)))
```

# Key Takeaways

- ① **Directional Derivatives** characterize behavior of convex functions at a point
- ② **Convexity Characterizations** provide first-order conditions via monotonicity of gradients
- ③ **Gâteaux vs Fréchet Differentiability:**
  - Gâteaux: Linear approximation in each direction
  - Fréchet: Uniform linear approximation
  - Equivalent in finite dimensions
- ④ **Subdifferentials** generalize gradients to non-smooth functions
- ⑤ **Continuity:** Differentiability on interior implies continuity
- ⑥ **Applications:** Optimization, monotone operators, descent methods

## Further Reading

- Proposition 17.2-17.4: Directional derivative properties
- Proposition 17.7-17.10: Convexity and strict convexity characterizations
- Proposition 17.21-17.22: Descent directions and steepest descent
- Proposition 17.31-17.41: Gâteaux and Fréchet differentiability
- Proposition 17.48-17.51: Continuity results
- Examples 17.8, 17.11, 17.13, 17.25, 17.33, 17.36, 17.46, 17.49: Important concrete examples